

MACHINE LEARNING-BASED AUTO INSURANCE FRAUD DETECTION: ANALYZING CUSTOMER, POLICY, AND INCIDENT RISK PATTERNS

by

Meshwa Patel
Student ID: 501390663

Master of Science, Data Science & Analytics

A Major Research Project presented to Toronto Metropolitan University
in partial fulfillment of the requirements for the degree of Master of Science
in the Program of Data Science and Analytics

Supervisor: Dr. Shengkun Xie

Toronto, Ontario, Canada, 2026

**AUTHOR'S DECLARATION FOR ELECTRONIC SUBMISSION OF A MAJOR
RESEARCH PROJECT (MRP)**

I hereby declare that I am the sole author of this Major Research Project. This is a true copy of the MRP, including any required final revisions.

I authorize Toronto Metropolitan University to lend this MRP to other institutions or individuals for the purpose of scholarly research.

I further authorize Toronto Metropolitan University to reproduce this MRP by photocopying or by other means, in total or in part, at the request of other institutions or individuals for the purpose of scholarly research.

I understand that my MRP may be made electronically available to the public.

Meshwa Patel

ABSTRACT

Machine Learning-Based Auto Insurance Fraud Detection
Meshwa Patel, Master of Science, 2026
Data Science and Analytics, Toronto Metropolitan University

Auto insurance fraud costs insurers a growing sum each year, and that cost eventually lands on legitimate policyholders through higher premiums. This Major Research Project builds and critically evaluates a supervised machine learning pipeline for identifying fraudulent auto insurance claims from structured customer, policy, and incident data, treating explainability as a first-class requirement rather than an afterthought. Four classifiers from distinct model families - a Logistic Regression, a Random Forest, an XGBoost, and a Multi-Layer Perceptron neural network - were trained on a corpus of 30,000 confirmed fraudulent and non-fraudulent auto claims (class imbalance of approximately 7.7 to 1), both with and without the Synthetic Over-sampling Technique (SMOTE), giving eight configurations in total. These eight models were compared under identical preprocessing and judged on precision, recall, F1-score, and AUC-ROC rather than accuracy, which this project shows actively misleads under heavy imbalance. The top performer, Logistic Regression with SMOTE, reached a test AUC-ROC of 0.726 and recall of 0.69, and Repeated Stratified K-Fold cross-validation confirmed these results held up. One finding runs counter to expectation: the more powerful neural network performed worst of the four. Together with a saturating learning curve and a below-chance transfer result, this points to the same conclusion: performance is limited by a weak, and possibly synthetically derived, dataset rather than by any shortcoming in the modeling approach. Model decisions were readily explained through SHAP and a transparent, data-driven Fraud Risk Score, with empirical fraud rates rising from 8.8% in the Low tier to 26.7% in the High tier. A fairness audit identified age as a biased attribute in the model and showed that the bias can be removed at negligible cost to performance simply by excluding demographic variables. The pipeline reached up to 0.855 test AUC-ROC on two independent, real-world datasets, which confirms the method is sound and puts its performance in line with related published work. Viewed as a triage and ranking tool rather than an automatic decision-maker, a comparatively simple machine learning model delivers really practical and monetary benefits.

Keywords: insurance fraud detection, supervised learning, class imbalance, SMOTE, XGBoost, neural networks, explainable AI, SHAP, algorithmic fairness, risk scoring, external validation

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. Shengkun Xie, for his guidance, insight, and constructive feedback throughout every stage of this Major Research Project. I am also grateful to the faculty of the Data Science and Analytics program at Toronto Metropolitan University for the foundation this work builds upon, and to the maintainers of the open-source scientific Python ecosystem, in particular scikit-learn, XGBoost, imbalanced-learn, and SHAP, without which this research would not have been possible.

CONTENTS

1	INTRODUCTION	1
1.1	Background and Motivation	1
1.2	Problem Statement and Research Questions	1
1.3	Scope, Contributions, and Structure	2
2	LITERATURE REVIEW	3
2.1	From Manual Review to Data-Driven Detection	3
2.2	Model Families for Structured Insurance Data	3
2.3	Class Imbalance and the Ambiguous Role of Resampling	4
2.4	Identified Gaps and Positioning of This Work	5
3	DATA DESCRIPTION AND EXPLORATORY ANALYSIS	7
3.1	Data Source and Structure	7
3.2	Key Features	7
3.3	Dataset Characteristics	8
3.4	Data Cleaning	9
3.5	Target Distribution	9
3.6	Univariate Analysis of Fraud Indicators	10
3.7	Correlation Structure	11
4	METHODOLOGY	13
4.1	Overview of the Analytical Workflow	13
4.2	Feature Engineering	13
4.3	Experimental Design and Data Partitioning	14
4.4	Treatment of Class Imbalance	14
4.5	Choice of Models	14
4.6	Evaluation Metrics	15
4.7	Explainability and Risk Scoring Methodology	15
5	RESULTS AND DISCUSSION	17
5.1	Model Comparison and the Effect of SMOTE	17
5.2	Performance of the Selected Model	19
5.3	Stability of Results under Cross-Validation	20
5.4	Model Interpretability	21
5.5	Fraud Risk Scoring	23
5.6	Fairness Audit and Mitigation	24
5.7	External Validation	25
5.8	Advanced Analyses	29
5.9	Synthesis and Discussion	31
6	CONCLUSION AND FUTURE WORK	33
	References	34

LIST OF FIGURES

3.1	Distribution of the target variable, showing the 88.5% to 11.5% class imbalance. . .	10
3.2	Fraud rate by incident features. Incident severity provides the clearest single separation, though the effect remains modest.	10
3.3	Distribution of claimant age for fraudulent and legitimate claims.	11
3.4	Correlation heatmap of numeric features. The near-absence of strong correlations is itself an important finding.	12
4.1	High-level project methodology.	13
5.1	Effect of SMOTE on recall and F1-score across the four models.	18
5.2	Confusion matrices for all eight configurations on the test set.	18
5.3	ROC and Precision-Recall curves for the selected model on the test set.	20
5.4	Cross-validated F1-score with standard-deviation error bars.	21
5.5	SHAP global feature importance for the selected model.	22
5.6	SHAP waterfall plot explaining a single high-risk claim.	23
5.7	Empirical fraud rate by risk tier.	24
5.8	Empirical fraud rate versus model flag rate by demographic group, prior to mitigation.	25
5.9	Flag rates by age group before and after removal of demographic features.	25
5.10	ROC curves for the independent benchmark on the fraud_oracle dataset.	27
5.11	ROC curves across all four datasets, each retrained in-domain. Two real datasets approach an AUC of 0.85, while one carries almost no signal.	28
5.12	Summary of AUC across all external datasets and test types.	28
5.13	Learning curve. Test AUC plateaus, indicating that additional data would not improve performance.	31

LIST OF TABLES

3.1	Key features of the dataset, grouped by category.	7
3.2	Summary of dataset characteristics.	9
5.1	Validation performance of all eight model configurations. The best configuration by F1-score is highlighted in bold.	17
5.2	Test-set performance of the selected model (Logistic Regression with SMOTE). . .	19
5.3	Cross-validated results (mean $\pm$ standard deviation) for the SMOTE models. . . .	20
5.4	Empirical fraud rate by data-driven risk tier on the test set.	24
5.5	Summary of external validation results. A transfer test applies the main-trained model directly; an in-domain test retrains on the target dataset.	26

1. INTRODUCTION

1.1 Background and Motivation

Insurance fraud is one of the costliest problems that the property and casualty insurance sector faces and it remains stubbornly difficult to solve. In the auto business insurance fraud ranges from a small opportunistic exaggeration on a real claim to a fully staged crash and even to the actions of organized criminal groups that involve many policyholders, claimants, witnesses and service providers. Industry studies have long estimated the cost to be in billions of dollars. Because insurers must pass that cost on to customers in order to stay in business the cost falls directly on policyholders through their auto premium rates. Insurance fraud detection is therefore a problem but at its core it is also a matter of basic fairness for all insured motorists.

The main difficulty is that a fake claim by its nature looks very much like a real claim. A person who is trying insurance fraud will give evidence: a consistent story supported by calculations that seem reasonable at first glance. Add to this the number of motor claims that even a moderately sized modern insurer processes and checking every file in detail by hand becomes economically impossible. Because of this detection has largely depended on a mix of human claims adjusters and rule-based engines that hand off a claim to an adjuster once it passes a set threshold. These methods share the drawbacks: they are slow and use a lot of resources when scaled up and because their logic is fairly standard and often publicly known a smart fraudster can bypass them.

A machine-learning solution takes a different approach. Of hand-coding the logic a supervised learning algorithm trains on a large set of claim histories where each case is labeled “fraudulent” or “not fraudulent” and it discovers on its own the statistical patterns that separate the two groups. After training the model rates how likely an incoming claim is to be fraudulent, and an insurer can focus its investigative resources on the claims that are most likely to pay off. This project uses that approach for a dataset on auto insurance, but it does not consider raw predictive performance as the only goal. Because labeling a claim means charging a customer, a system that explains why a claim was flagged is important if its results are to be used by an investigator and defended as fair.

1.2 Problem Statement and Research Questions

The central research question this project addresses is: can a machine learning model accurately identify fraudulent auto insurance claims from structured claims and policyholder data, and can its decisions be explained clearly enough to support a human investigator?

This overarching question breaks down into several concrete sub-questions that structure the investigation:

- (i) Among four classifiers drawn from distinct model families, which offers the best balance of predictive performance and interpretability on this problem?
- (ii) By how much, and in which direction, does synthetic oversampling with SMOTE change

each model’s precision, recall, and F1-score relative to training on the original imbalanced data?

- (iii) Which features does the best model rely upon most heavily, and are these consistent with domain intuition about fraud indicators?
- (iv) Can we group claims into three categories—high, medium, and low risk—in a way that makes sense and is consistent with the actual outcomes and the predictions made by our model? We want to make sure that the claims in each category have fraud rates that match up with the true labels and the probabilities predicted by the model.
- (v) Does the top-performing model treat different groups of people unfairly, like older or younger people, men and women, or those with more or less education? And if it does, can we fix this problem without making the model work a lot worse?
- (vi) Does the methodology generalize beyond the primary dataset, and what does its behaviour on independent datasets reveal about the nature of the primary data itself?

1.3 Scope, Contributions, and Structure

Consequently, the scope of this project purposely covers only structured, tabular data claims; the inclusion of unstructured data such as adjuster notes or photos has been reserved for a later phase. Within this scope, this project makes four contributions that can help to bridge the literature gaps mentioned earlier:

- (1) a like-for-like study of four models under the same conditions;
- (2) the estimation of the effect of SMOTE, rather than taking its positive impact for granted;
- (3) a two-level explanation approach that combines the SHAP technique and English-language risk scoring;
- (4) an external validation test to distinguish between methodology and data issues.

The rest of this report is structured in the following way. In Section 2 we review the literature related to this study. Section 3 deals with data and exploratory analysis which provided guidance for modelling. Section 4 gives the full details of the methodology. The results are reported in Section 5, including model comparison, cross-validation, interpretability, fairness, external validation and some additional analyses. Finally, Section 6 concludes with some recommendations for further research.

2. LITERATURE REVIEW

2.1 From Manual Review to Data-Driven Detection

Insurance fraud has been a longstanding issue, existing on a spectrum ranging from the inflation of claims made to making a claim when nothing of the sort occurred. Viaene and Dedene (2004) offered one of the earliest approaches to the study of fraud as a business problem, and they include a detail that is still relevant – the detection of fraud is adversarial since the group of fraudsters evolves in response to the controls imposed upon them. The adversarial nature of the process can thus be used as justification for the non-rigid approach to detecting fraud. Brinkmann (2005) poses a similar question from a different point of view: why do honest customers lie when reporting claims? This approach considers fraud as an opportunity issue, and rationalization, which explains the difficulty of spotting the statistical signal in the data on claims. Aslam et al. (2022) present a contemporary state of the art regarding fraud detection.

It wasn't an instant change. One of the most widely cited papers on techniques for mining data in order to detect financial fraud came out in 2011, authored by Ngai, Hu, Wong, Chen & Sun. The authors list techniques ranging from logistic regression to decision trees and neural networks along with types of fraud they can be used against – and that's another reason logistic regression continues to be a reasonable baseline to use rather than an overly simplistic technique to brush off. From the insurance industry perspective, Kemp (2010) provides this practitioner's view of how the number of claims had become too high to handle before there were even any automated methods available. There is the same basic problem pattern found outside the insurance industry too – predicting loan default (Liu et al., 2022), and corporate solvency (Abdullah, 2021).

2.2 Model Families for Structured Insurance Data

In their research of different machine learning algorithms applied to microdata on property insurance, Severino and Peng (2021) established that tree-based ensemble algorithms work particularly well when dealing with structured claim data in tables, thus explaining the decision to employ Logistic Regression together with two tree-based ensembles (Random Forest and XGBoost) in the study described below. Similarly, Owolabi, Shahra, and Basurra (2023) conducted an experiment with several classification models trained on auto insurance fraud datasets. The authors' claim that comparison is itself an important methodological point of view has been acknowledged.

Fraud detection on cards is a more developed field, and a few ways of modeling rare events are based on it. Adewumi & Akinyelu (2017) review machine learning and nature-inspired techniques for credit-card fraud analysis, while Carneiro, Figuera, and Costa (2017) provide a case study of a data mining system deployed in a live e-commerce environment, which provides some information about how these techniques work when deployed. Network-based features were introduced to this field by Van Vlasselaer et al. (2015) through tools like APATE, and Forough & Momtazi (2021) and Zhang et al. (2021) both attempts to tackle card fraud with deep neural networks or other heavy

modeling approaches. Correlation-based feature selection [5] is similar to the pruning approach employed here. Sequence- and network-based heavier models have been considered but rejected in favor of lighter models because of the relatively small size of the database (thousands, not millions of observations).

The next category of studies considers deeper into the unstructured or relational data associated with insurance claims. Wang & Xu (2018) employ deep learning along with topic modeling techniques for interpreting the free-text justifications in the auto claims' reports, which seems quite promising because these narratives may contain signal not captured by coded data fields. Van Vlasselaer et al. (2017) base their social security fraud detection system, "GOTCHA", on the network analysis, using the relational approach which becomes even more powerful when policyholders are connected in the network. However, these approaches cannot be applied in the current project because there is neither claim text nor a policyholder network among the variables in this dataset, so these areas can become promising for future research. The last category is associated with situations where fraud labels are not available. For example, Nian et al. (2016) and Gomes et al. (2021) apply unsupervised spectral ranking and unsupervised deep learning to auto insurance claims respectively, but it seems less relevant to the current project because of the presence of the accurate fraud label in the used dataset.

2.3 Class Imbalance and the Ambiguous Role of Resampling

This class imbalance problem pervades the entire body of literature mentioned above because in practice, valid entries are present in 10 times or more numbers compared to invalid entries. This means that predicting "no fraud" always will give a model very good accuracy despite being completely useless in practice. The common solution to this problem is to use oversampling on the training set, mostly via the technique called SMOTE which creates synthetic entries from existing fraud cases by interpolating them. Komsrimorakot and Siriborvornratanakul (2025) do so quite successfully in an automobile insurance context. They suggested using a combination of center point SMOTE and random undersampling and tested their method on three open datasets. According to them, recall was raised every time while precision dropped a little bit, although they emphasize that this should not be considered as a universal tendency. The results obtained in this project follow a similar trend: SMOTE helped to increase recall but did not affect AUC-ROC much, lying somewhere between the benefit and the damage inflicted by other researchers on other models.

A model's prediction should never be the final word on whether to pursue a claim. Every flagged claim still needs a human investigator to decide whether it gets paid, denied, or sent back for more documents, and denying a claim on the strength of an unexplained score alone raises real problems of fairness and legibility. Ekici (2025) applies SHAP (SHapley Additive Explanations) to auto insurance fraud models to produce ranked, per-claim feature explanations, and Carmona, Dwekat and Mardawi (2022) use an XGBoost classifier with a similar SHAP-based method to analyze business failure. Both studies make a strong case for SHAP, but in each case the explanation it produces is really an artifact for an audience that already understands data science, not a tool an investigating officer could use directly. That gap is what the two-stage explanation system in this project was built to close: a model-based explanation through SHAP, paired with a model-independent Fraud Risk Score from a rule-based system (Section 4.7) that a non-technical analyst

can read on its own.

Four more recent articles have come out that are sufficiently similar to this project to be compared directly. Ding et al. (2025) use particle swarm optimization to optimize parameters in XGBoost and apply this method to an auto-insurance fraud detection data set; the optimized model obtains up to 95% accuracy in prediction, 3% higher than a standard parameter XGBoost. Logistic regression, backpropagation NN, and Random Forest scored 54.1%–68.6%, and 78.4% respectively. Additionally, according to their SHAP analysis, Claim Amount and Vehicle Type are the two factors with the highest influence on XGBoost predictions. Interpreted this way, this article is a caution about over-reliance on an optimized model: such an approach introduces additional computation time and makes it difficult to replicate results or transfer the model to another data set. This is why this project compares several models under normalized conditions.

While Ekici (2025) emphasizes the importance of explainability from the start, using global SHAP feature importance along with the waterfall plot per claim, he uses SHAP along with logistic regression coefficients and p-values for statistical inference. My replication is done by using the waterfall plot directly as the explanation for each claim for claims investigators. The effect of class weighting or SMOTE is confirmed in my study as well, just as Ekici discovered that it can be problematic when the metric used is not capable of dealing with changes in the class distribution. Komsrimorakot and Siriborvornratanakul (2025) give the most serious consideration to the issue of class imbalance of all the papers discussed, conducting multiple class balancing methods for motor insurance claims data. My replication is based on the exact same procedure with identical preprocessing done, with each model being run with and without SMOTE. Dondapati (2025) conducts experiments on classification models for car insurance claim data, arguing that model comparison is better than guessing one model.

2.4 Identified Gaps and Positioning of This Work

Reading across this body of work, four gaps recur, and each became a deliberate design choice in this project. First, genuine like-for-like model comparison, holding preprocessing and data splits constant, is rarer than the volume of published work suggests; many studies tune a single model in isolation, as Ding et al. (2025) do, which risks conflating the effect of the model family with the effect of tuning effort. This project compares four models under identical preprocessing and identical data splits. Second, the effect of resampling is often asserted rather than measured: much of the imbalance literature, including Komsrimorakot and Siriborvornratanakul (2025), reports only the SMOTE-treated result. This project trains every model with and without SMOTE specifically to quantify that effect rather than assume it. Third, explainability, where it appears, often stops at the data scientist rather than reaching the front-line investigator, as in Ekici (2025) and Carmona et al. (2022); this project pairs SHAP with a transparent, rule-based score that needs no machine learning infrastructure to use. Fourth, honest reporting of modest results on difficult, possibly synthetic data is uncommon in literature that tends to emphasize high headline accuracy; this project reports its results plainly and uses external validation on independent datasets to interpret them rather than read them in isolation.

After reviewing the literature for this paper, it seems clear that there are four fundamental and

common limitations in the field, which I have then specifically tried to design against here.

1. **Lack of true, like-for-like comparison.** In such a vast field, I thought it odd that I seemed unable to find many papers that truly benchmark one against the other on an equivalent dataset and processing method. It seems to be far more common that one model is optimized over another (as done by Ding et al. 2025) thereby confounding the difference between models and the effort involved. In this study, four different types of models were tried on exactly the same dataset splits and preprocessing steps.
2. **Availability of studies on the effect of resampling.** Most papers with a class imbalanced dataset seem to simply state the results once resampling has been performed on it, rather than examine the impact of that step (an approach used by Komsrimorakot and Siriborvornratanakul 2025). Instead, in order to be aware of the impact that SMOTE is having on the individual algorithms, SMOTE has been implemented with and without and then compared.
3. **Lack of reachability of explanation.** Where explainability has been studied, for example with the paper used of Ekici (2025) and Carmona et al. (2022) the explainability only gets to the data scientist. We, here are pairing machine learning explanations (e.g. SHAP) with a rule based score that anyone with domain knowledge could verify.
4. **No genuine report on tough data.** We saw very few reports of average accuracies on challenging datasets; most of these models offer their successes only when evaluated on relatively pristine datasets, we hope that reporting our average accuracies is a realistic step forward and as an independent measure of validity will also test upon several other challenging datasets.

When combined, all four gaps speak to the same underlying issue: so much of the published literature has optimized for a headline-generating number instead of for an answer an actual person would trust and be able to act upon. That decision was the driving force behind nearly every methodological decision in the analysis that followed, whether it meant testing four models simultaneously rather than seeking the highest score on a single one, taking the SMOTE method as something we had to examine rather than treat as a known quantity, or building an explanatory layer a claims investigator could understand in addition to one a data scientist could appreciate. It also makes clear why we took external validation on unseen, independent datasets as not just a nice feature but an essential ingredient in the analysis; a fraud-detection model that seems to do well solely against its training data isn't that much better than the one-and-done experiments we are seeking to transcend. The four gaps were less a checklist in that light than a practical definition of what any sound fraud-detection study would have to achieve.

3. DATA DESCRIPTION AND EXPLORATORY ANALYSIS

3.1 Data Source and Structure

This research draws on the Car Insurance Fraud Dataset, a collection of auto insurance claim records from the United States. The dataset contains 30,000 records spanning November 2023 to November 2025, with a confirmed binary fraud label (`fraud_reported`: Y = Fraud, N = Not Fraud) for every record.

Dataset Link: <https://www.kaggle.com/datasets/ahluwaliasaksham/car-insurance-fraud-detection-dataset>

The dataset contains 30,000 observations, each representing a single auto insurance claim. It covers several dimensions:

3.2 Key Features

Table 3.1 summarizes the dataset's fields by category.

Table 3.1: Key features of the dataset, grouped by category.

Category	Feature	Description
1. Policy Features	<code>policy_id</code>	Unique policy identifier
	<code>policy_state</code>	US state where the policy was issued (10 states: CA, FL, GA, IL, MI, NC, NY, OH, PA, TX)
	<code>policy_deductible</code>	Deductible amount (\$200–\$1,000)
	<code>policy_annual_premium</code>	Annual premium paid by the insured
2. Customer Features	<code>insured_age</code>	Age of the insured person
	<code>insured_sex</code>	Gender of the insured (Male / Female / Other)
	<code>insured_education_level</code>	Education level (High School / College / Masters / PhD)
	<code>insured_occupation</code>	Occupation category (Doctor, Lawyer, Sales, Clerk, Engineer, Manager, Teacher, Technician)

(continued on next page)

(continued from previous page)

Category	Feature	Description
	insured_hobbies	Hobby category (7 types including camping, chess, yachting, hiking)
3. Incident Features	incident_date	Date of the incident (November 2023 to November 2025)
	incident_type	Type of incident (Multi-vehicle Collision, Single Vehicle Collision, Parked Car, Vehicle Theft)
	collision_type	Collision direction (Front, Rear, Side, Unknown)
	incident_severity	Damage level (Minor Damage, Major Damage, Total Loss)
	authorities_contacted	Emergency service contacted (Police / Ambulance / Fire / None); 25.2% missing
	incident_state, incident_city	Geographic location of the incident
	incident_hour_of_the_day	Hour of the incident (0–23)
	number_of_vehicles_involved	Count of vehicles in the incident
	bodily_injuries	Number of reported bodily injuries
	witnesses	Number of witnesses present at the scene
	police_report_available	Whether a police report was filed (Yes / No)
4. Claim Features	claim_amount	Individual claim amount filed
	total_claim_amount	Total claim amount including all components
5. Target Variable	fraud_reported	Binary label: Y = Fraudulent claim, N = Legitimate claim

3.3 Dataset Characteristics

Table 3.2 summarizes the overall characteristics of the dataset.

Table 3.2: *Summary of dataset characteristics.*

Characteristic	Value
Type	Structured tabular dataset
Nature	Structured to resemble real-world auto insurance claims data – United States (later analysis in Section 5.9 finds statistical evidence that this dataset is likely synthetic)
Total Records	30,000
Total Features	24 (excluding target variable)
Fraud Cases	3,440
Non-Fraud Cases	26,560
Class Imbalance Ratio	7.7:1
Date Coverage	November 9, 2023, to November 8, 2025 (2 full years)
Geographic Coverage	10 US states – CA, FL, GA, IL, MI, NC, NY, OH, PA, TX
Missing Values	<code>authorities_contacted</code> column – 7,564 missing values (25.2%)
Duplicate Records	None

3.4 Data Cleaning

There was little cleaning to be done – the idea I will come back to next time when I discuss “What if it’s synthesized?” There was one feature, `authorities_contacted`, which had 25.2% nulls. Rather than just throwing away 1/4 of my data, I concluded that this feature means “not in contact with any of the officials” and encoded it as “None.” As it turned out, this proved to be one of the strongest predictors of fraud! The `incident_date` column was transformed into a datetime. All categorical textual features were made consistent by encoding categories consistently. The fraud label was numerically encoded from `fraud_reported`. The `policy_id` and `incident_city` columns were kept in the dataset but were not fed to the model as inputs in order to avoid memorization of incident reports.

3.5 Target Distribution

The most important feature of the data is the distribution of the target variable, shown in Figure 3.1. Because of the extreme imbalance, a naive classifier that predicts the majority class for every claim reaches 88.5% accuracy while identifying zero fraudulent claims. This is why, as discussed in Section 4.5, accuracy is not used here as an evaluation metric.



Figure 3.1: Distribution of the target variable, showing the 88.5% to 11.5% class imbalance.

3.6 Univariate Analysis of Fraud Indicators

Looking at each individual factor, we find that the correlation is much weaker and less clear. For example, “total loss” claims are approximately 14.8% fraudulent while “minor damage” claims are about 9.6% (see Figure 3.2). This is not a large difference, and the same trend applies to all other factors – small but consistent increases. The younger people making the claims commit slightly more fraud than the older (see Figure 3.3), there are slightly more fraudulent claims with witnesses, and so forth.

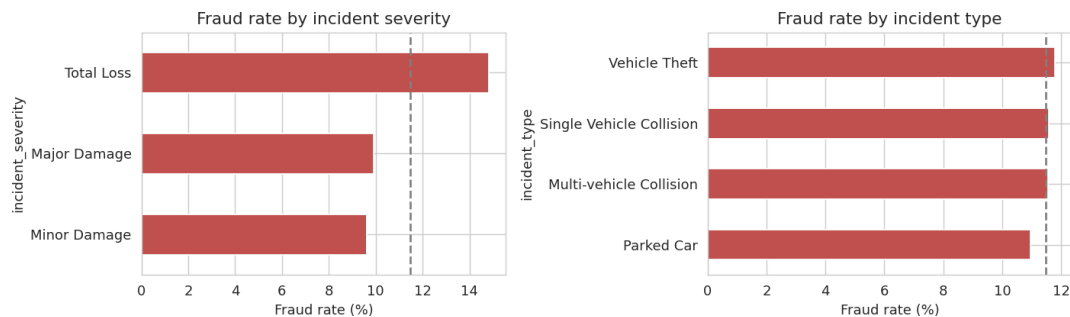


Figure 3.2: Fraud rate by incident features. Incident severity provides the clearest single separation, though the effect remains modest.

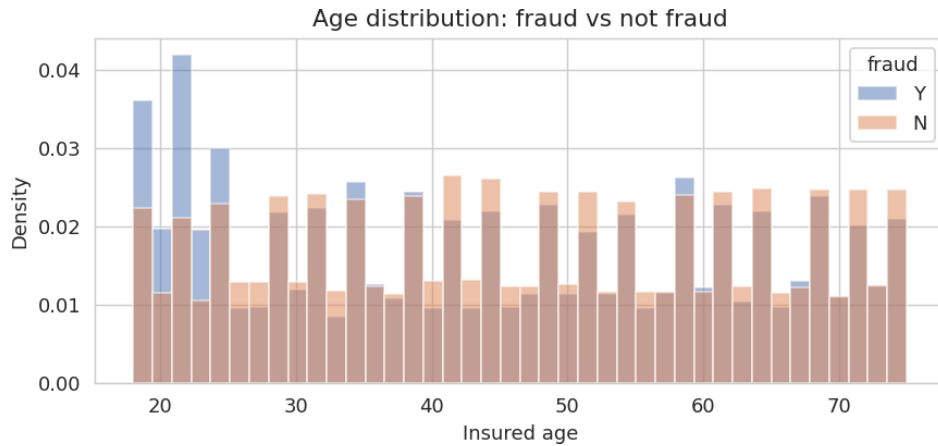


Figure 3.3: *Distribution of claimant age for fraudulent and legitimate claims.*

3.7 Correlation Structure

The weak univariate signal held up at the multivariate level too. Correlation analysis of the numeric features (Figure 3.4) shows that no feature correlates with the fraud target beyond a slight degree, and the features show little correlation with one another either, aside from cases of tautological dependence, such as individual claim amount and total claim amount. This is roughly what you'd expect from a problem where fraud has to be inferred from multiple weak signals rather than one strong one, and it hints early on at both the limited accuracy of the models that follow and the seemingly artificial nature of the dataset.

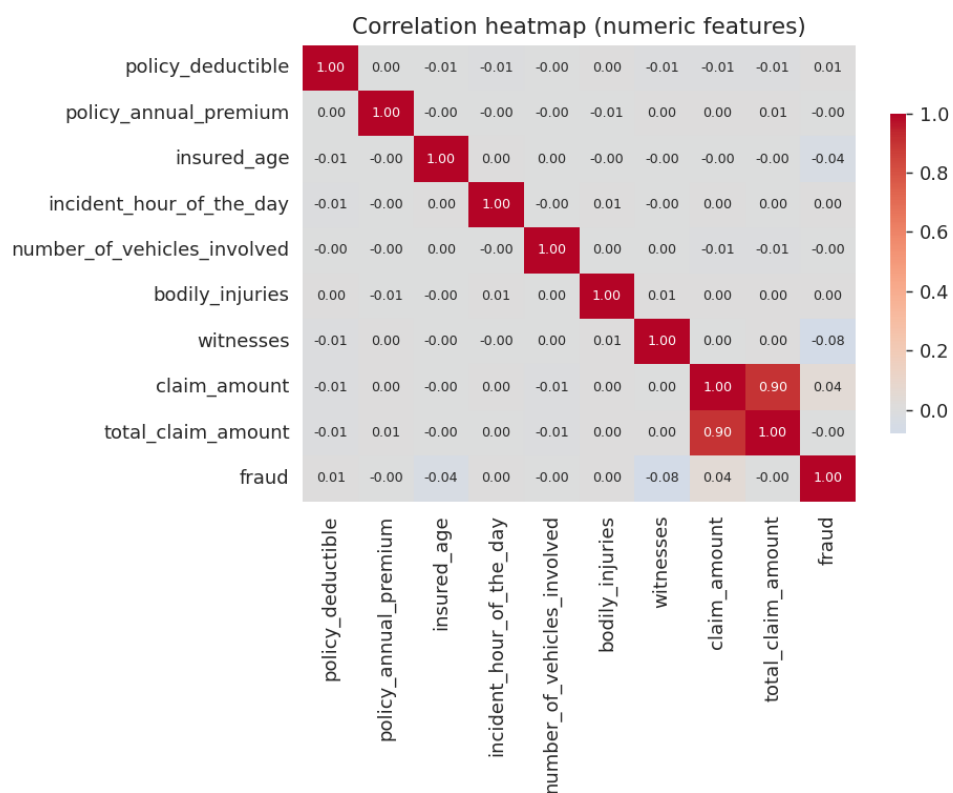


Figure 3.4: *Correlation heatmap of numeric features. The near-absence of strong correlations is itself an important finding.*

4. METHODOLOGY

4.1 Overview of the Analytical Workflow

The project follows a coherent, reproducible workflow from raw data through to a fully evaluated and interpreted model. Figure 4.1 shows this workflow at a high level and traces each step in sequence, from the original business problem through to the analytical methods used for modeling. Every stage was designed with the validity of the evaluation in mind, particularly to avoid the information leakage that so often inflates a fraud model's apparent performance.

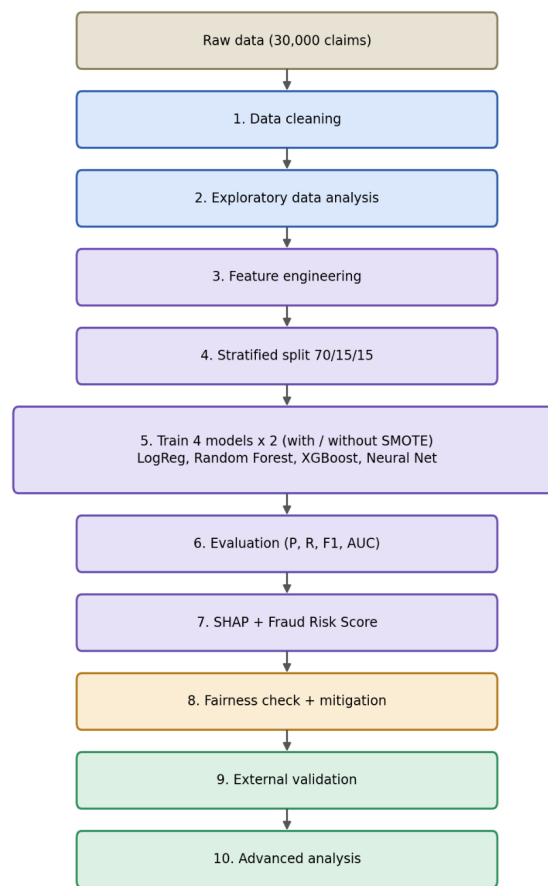


Figure 4.1: *High-level project methodology.*

4.2 Feature Engineering

Because the individual raw features carried only weak signals on their own, I combined them into seven additional features capturing associations that don't show up directly in the raw columns.

Two are ratio features: `claim_to_premium = claim_amount / policy_annual_premium`, which controls for large individual claims relative to premium paid, and `total_to_claim_ratio = total_claim_amount / (claim_amount + 1)` (the +1 is a smoothing term to avoid division by zero), which controls for claims with an artificially high total value. Three are binary flags: absence of witnesses, absence of a police report, and no authority contacted. An ordinal `severity_score` variable was also defined, reflecting the natural ordering of incident severity (Minor Damage= 1, Major Damage= 2, Total Loss= 3). Two time-based features, incident month and incident weekday, were derived from the incident date field. The individual identifying label and the raw date/city columns were excluded; that exclusion is explained in Section 3.2.

4.3 Experimental Design and Data Partitioning

The generated dataset is then divided into three stratified partitions, in the ratio 70/15/15 (train, validation, and test sets correspondingly), in order to preserve the 11.5% fraud rate observed in the entire dataset. This division into three sets is required from the methodological point of view: the train set is used for estimating parameters of the chosen model, the validation set is used to evaluate and select among competing models, while the test set is held back and analyzed just once, at the very end, to obtain the unbiased performance estimate shown in Section 5.2. If the model selection had been performed using the test set, the performance estimate obtained would be biased positively because the selection process itself would affect the estimate. The entire preprocessing procedure takes place within the scikit-learn pipeline, meaning that the scaler and encoder are trained on the train set and applied consistently to all three datasets.

4.4 Treatment of Class Imbalance

The imbalance in question (7.7 to 1) was tackled using SMOTE (Synthetic Minority Oversampling Technique). In essence, SMOTE enlarges the training dataset by synthesizing additional minority class examples via interpolation of existing fraud cases, making sure there is enough exposure to the minority pattern for the classifier to learn from it. Instead of being applied beforehand to the whole dataset, the SMOTE technique was implemented within the modeling pipeline, ensuring that synthetic examples would be created only based on the training part of a particular split and not based on validation or test data. Preprocessing the whole dataset using SMOTE before the train/val/test split is often done as a workaround but results in leaking information and score boosting into the validation and test datasets, therefore it was not used in this case. Instead, the impact of resampling was verified experimentally, and each model was trained twice on each version of the data, resulting in eight model versions.

4.5 Choice of Models

Four classifiers have been selected carefully in order to demonstrate a variety of model architectures, rather than compare many variations of similar methods. Logistic Regression (`max_iter=1000`) is a simple and interpretable linear classifier with easy-to-understand coefficients.

Random Forest (`n_estimators=200`) and XGBoost (`n_estimators=300`, `learning_rate=0.1`, `max_depth=5`, `tree_method="hist"`) are representatives of two assembling techniques for bagging and boosting tree models which are able to learn non-linear dependencies in the dataset. A Multilayer Perceptron with two fully connected hidden layers of 32 and 16 neurons (`activation="relu"`, `solver="adam"`, `max_iter=300`, `early_stopping=True`) demonstrates one of the most effective and popular model architectures. A neural network should be included in the list because there is one specific purpose of including such a complex architecture – if it is not better than a simple linear classifier, the ceiling lies in the data. All four models were seeded with `random_state=42` for reproducibility, and, where SMOTE was applied, `SMOTE(random_state=42)` (default `k_neighbours=5`) was inserted between the preprocessor and the classifier inside an imbalance pipeline so that it only ever resampled the training fold.

As a robustness check on the model-selection decision, XGBoost was additionally tuned via `GridSearchCV` over `max_depth` $\in \{3, 5\}$, `learning_rate` $\in \{0.05, 0.1\}$, and scored on AUC-ROC with 3-fold stratified cross-validation. The best configuration (`max_depth=3`, `learning_rate=0.05`, `n_estimators=400`) reached a cross-validated AUC of 0.756 and, when evaluated once on the test set, an AUC of 0.770 – higher than both the untuned XGBoost model and the selected Logistic Regression with SMOTE (0.726). This does not overturn the model-selection decision, since the selection criterion was validation F1-score rather than AUC and Logistic Regression with SMOTE remains the strongest model on that metric, but it shows that a properly tuned XGBoost is a competitive alternative if AUC-ranking, rather than F1-balanced classification, is the operational priority.

4.6 Evaluation Metrics

The large imbalance made the simple accuracy metric meaningless, and thus the evaluation of the models was based on metrics appropriate for the classification of rare events. Precision, or Positive Predictive Value (PPV), is the ratio of the cases flagged as fraudulent to the total number of flagged cases. It shows how often the accusation of fraud will be true. Recall, or hit rate, or sensitivity, is the ratio of the actual frauds detected to the total number of actual frauds. It shows how often fraud will be detected. F1-score is the harmonic mean of precision and recall, and it served as the most stable measure for comparison of the models. AUC-ROC shows the discrimination power of the model independently of the chosen threshold.

4.7 Explainability and Risk Scoring Methodology

Interpretability was considered from two perspectives. From the model-level perspective, the selected model was used to create an overall ranking of the features' importance, while SHAP was used to interpret individual claims. Because the winning model was Logistic Regression, a Linear Explainer was used (tree-based models, when examined, used Tree Explainer, and the neural network used Kernel Explainer with a 30-point `shap_kmeans` background summary for tractability). SHAP values were computed for a random sample of 500 test claims (100 for the neural-network case, for speed), and two local waterfall explanations were produced for the

highest-predicted-probability claims in that sample.

From the operational perspective, a rule-based Fraud Risk Score was generated so that investigators would be able to work without having to interact with the model directly. The weights for this score were not manually assigned but were rather generated empirically: seven candidate red flags were each scored for their fraud lift (the flag's fraud rate divided by the 11.47% training base rate), a flag was retained only if its lift was at least 1.10, and the retained flags' points were then rescaled so that the strongest flag was worth 30 points. Tier cut-points (Low/Medium/High) were set from the 45th and 85th percentiles of the resulting score distribution on the training set.

5. RESULTS AND DISCUSSION

5.1 Model Comparison and the Effect of SMOTE

The results of applying all 8 models to the validation set are shown in Table 5.1. There are two main points worth noting. The first is a significant difference between the recall of all models trained on the imbalanced data and those trained with the application of SMOTE. Namely, all models that work without SMOTE show recall close to zero, implying that the model is almost not making any effort to detect fraud – recall for logistic regression trained on the unbalanced data is only 1.6%. Using SMOTE while training increases recall to 0.612 from 0.016 on the unbalanced data. The second point is more subtle and can be observed from the column containing AUC values. SMOTE’s effect on AUC is small and, for most models, is not an improvement: AUC is essentially unchanged for Logistic Regression (0.698→0.695) and XGBoost (0.731→0.731), and actually decreases slightly for Random Forest (0.740→0.736). The one exception is the Neural Network, whose AUC rises substantially under SMOTE (0.603→0.654, the largest AUC change in the table). What SMOTE mainly does, for the tree- and linear-based models at least, is change the threshold of commitment to the fraud classification when the model is confident enough, rather than improve the model’s underlying discriminative power. This is broadly consistent with Ekici (2025), though the neural network case shows the relationship is not universal across model families.

Table 5.1: Validation performance of all eight model configurations. The best configuration by F1-score is highlighted in bold.

Model	Data	Precision	Recall	F1	AUC-ROC
Logistic Regression	Original	0.364	0.016	0.030	0.698
Logistic Regression	SMOTE	0.207	0.612	0.310	0.695
Random Forest	Original	0.333	0.004	0.008	0.740
Random Forest	SMOTE	0.434	0.045	0.081	0.736
XGBoost	Original	0.440	0.099	0.161	0.731
XGBoost	SMOTE	0.431	0.103	0.166	0.731
Neural Network	Original	0.000	0.000	0.000	0.603
Neural Network	SMOTE	0.218	0.357	0.271	0.654

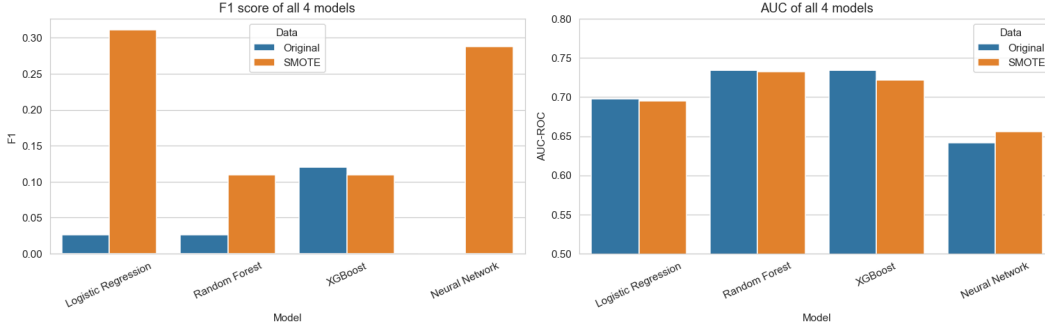


Figure 5.1: Effect of SMOTE on recall and F1-score across the four models.

The model selected by validation F1-score was Logistic Regression with SMOTE. Its F1 of 0.310 is nearly double the next best configuration, and it carries the added advantage of being the most interpretable model in the comparison. The neural network's performance tells its own story: it posted the lowest AUC of all models at 0.654, and in its original configuration it did not classify a single fraudulent claim. That the most expressive model here loses so badly to a simple linear one is not a failure of implementation. It is a finding in its own right: the data does not contain the kind of complex, non-linear structure a neural network is built to exploit. Figure 5.2 shows the confusion matrices for all configurations and makes the precision-recall trade-off concrete for each model.

The lesson learned here is quite simple: the simpler and more easily interpreted model won out over the more complex one for this particular dataset. This is not a mistake in implementation but rather an example provided by the data that, however correlations may lie between these variables and fraud, they are linear and there are no non-linear structures to discover. The confusion matrices in Figure 5.2 illustrate the same point in another way: recall and precision vary greatly depending on the configuration, and the neural network row is almost empty.

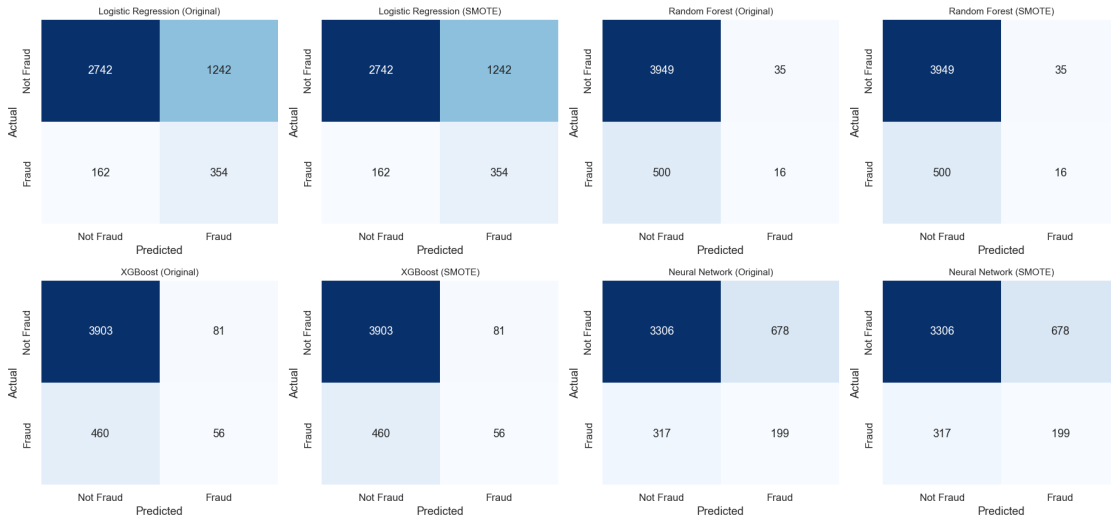


Figure 5.2: Confusion matrices for all eight configurations on the test set.

5.2 Performance of the Selected Model

This model was run once using the holdout test dataset containing 4,500 claims, as seen in Table 5.2. Since there is a close agreement between the test and validation F1 scores at 0.335 and 0.310 respectively, the model does not seem to be overfitted and can be trusted with data from the larger set. In production, this model will label 1,600 out of the total 4,500 claims submitted and will be able to detect 354 out of the 516 real cases of fraud among the labeled claims. A high recall coupled with low precision is what you expect of a tool meant to do risk triaging, and not fraud determination. See Figure 5.3.

Error analysis. Of the 516 real frauds in the test set, 354 were caught and 162 were missed, alongside 1,242 false alarms. The missed frauds systematically carry fewer red flags than the caught ones – only 12.3% of missed frauds have the no-witness flag versus 40.7% of caught frauds, and 47.5% versus 54.0% for the no-police-report flag – which suggests the model is not failing randomly but is specifically missing frauds that don't look like the frauds it was trained to recognize. A notable fraction of the misses are near-misses: 42 of the 162 missed frauds (26%) scored between 0.40 and 0.499, just below the 0.50 classification cutoff, with an average predicted probability of 0.327 among all missed frauds.

Threshold analysis. The default 0.50 threshold used throughout this chapter is not the F1-optimal one. Sweeping thresholds from 0.30 to 0.75 on the validation set shows the best F1 is actually achieved at a threshold of 0.60, giving test-set $F1 = 0.362$ and $\text{recall} = 0.548$ with 1,047 claims flagged, versus $F1 = 0.335$ and $\text{recall} = 0.686$ with 1,596 claims flagged at 0.50. The two operating points trade recall for precision and represent genuinely different deployment choices rather than one being objectively correct: 0.50 catches more fraud at the cost of more false alarms, while 0.60 is more conservative. A cost-based selection, sweeping the ratio of the cost of a missed fraud to the cost of a false alarm over $C \in \{3, 5, 10, 20\}$, yields optimal thresholds of 0.74, 0.58, 0.52, and 0.29 respectively – reinforcing that the "right" threshold depends on the business's relative tolerance for missed fraud versus false alarms, and the 0.50 default used for headline reporting in this chapter is only one reasonable choice among several.

Table 5.2: Test-set performance of the selected model (Logistic Regression with SMOTE).

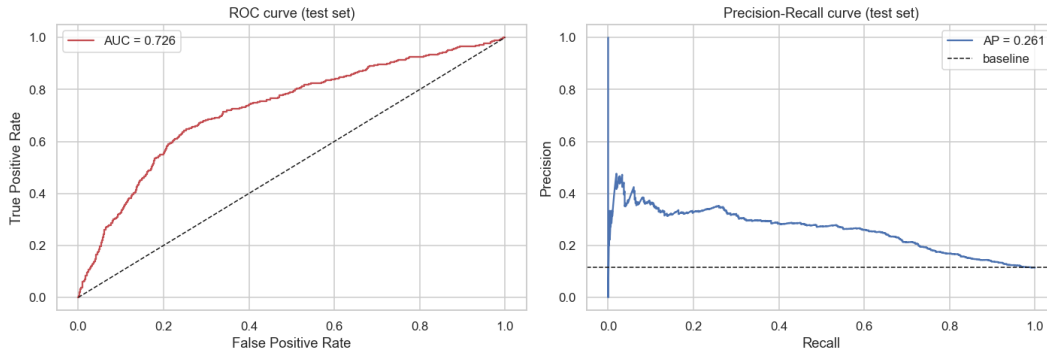
Metric	Value
Precision	0.222
Recall	0.686
F1-score	0.335
AUC-ROC	0.726

There are three main observations. First, the standard deviations of the measures are small, around 1 percent or even less, meaning that the findings can be seen as reliable and not just an outcome of randomness. Second, the highest F1 score is produced by Logistic Regression with SMOTE: it amounts to 32.6%. The 95% confidence interval of Logistic Regression with SMOTE (32.1%, 33.0%) does not intersect with the intervals of any other classifier, which means that the lead is statistically significant. Third, and most importantly, all classifiers excluding SMOTE have an

Table 5.3: *Cross-validated results (mean $\pm$ standard deviation) for the SMOTE models.*

Model (SMOTE)	Accuracy %	F1 %	AUC %
Logistic Regression	69.3 $\pm$ 0.5	32.6 $\pm$ 0.6	71.2 $\pm$ 0.6
Neural Network	79.7 $\pm$ 0.9	30.3 $\pm$ 0.7	67.3 $\pm$ 0.6
XGBoost	87.9 $\pm$ 0.2	18.0 $\pm$ 1.2	75.1 $\pm$ 0.9
Random Forest	88.2 $\pm$ 0.2	7.9 $\pm$ 1.2	75.1 $\pm$ 0.7

accuracy of less than 88% – while predicting zero fraud for all transactions will amount to 88.5% accuracy. One caveat worth noting: F1 was the selection criterion used throughout this project, and Logistic Regression with SMOTE clearly wins on F1, but it does not lead on cross-validated AUC – Random Forest (Original) tops the AUC ranking at 75.5%, ahead of Logistic Regression with SMOTE’s 71.2%. This is consistent with the earlier finding (Section 5.1) that SMOTE mainly reshapes the decision threshold rather than the underlying ranking ability of a model, and it confirms that the choice of evaluation metric materially affects which model looks best.

**Figure 5.3:** *ROC and Precision-Recall curves for the selected model on the test set.*

5.3 Stability of Results under Cross-Validation

To check if these results were not just a consequence of some specific train-test partitioning, we repeated our experiment by applying Repeated Stratified K-Fold cross-validation with 5 folds and 2 repeats (10 runs per configuration). The standard deviations in this case were low, usually not exceeding 1%. The 95% confidence interval of the F1 score of the selected model does not overlap with the confidence interval of the next best model; therefore, there was no luck in choosing the model. Cross-validation also gave us the answer regarding the appropriate metric – for all the models except those with the use of SMOTE, accuracy was around 88%, only slightly better than the classifier that predicts the majority class all the time (the accuracy of such a classifier is 88.5%). Obviously, accuracy is not a proper metric for this task. Figure 5.4 presents the cross-validated F1 score of each configuration and the corresponding standard deviation.

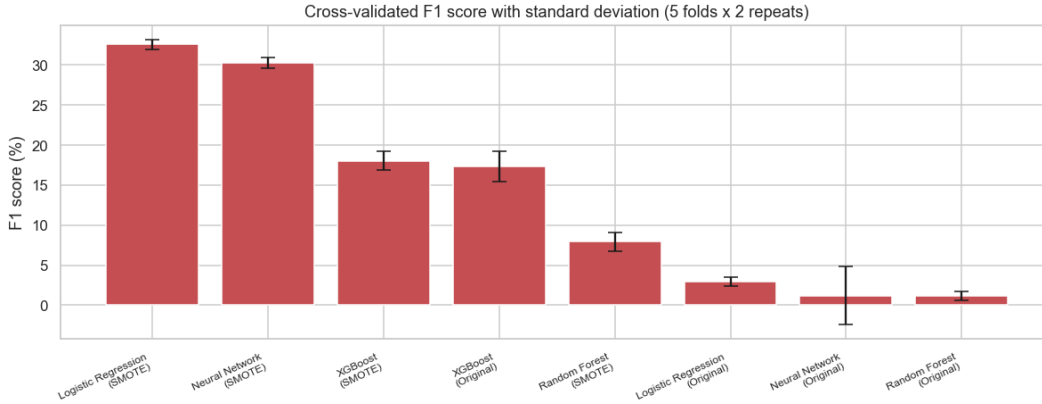


Figure 5.4: *Cross-validated F1-score with standard-deviation error bars.*

Explanation of why the cross-validation performance is stable: This stability is achieved through a mechanical process rather than luck. SMOTE was only used in the training folds and never in the validation fold; hence, no minority-class synthetic instance got to be part of the data being evaluated against the model. The F1 and AUC scores are thus accurate since they reflect true out-of-fold estimates that have not been boosted by synthetic instances that should not be there in the first place. In addition, having two repetitions of 5-way folds (thus 10 runs per model), each with different random splits, smooths out sampling variability.

Reproducibility of the cross-validation: for reproducibility, we used scikit-learn’s Repeated Stratified K-Fold with `n_splits=5`, `n_repeats=2`, and `random_state=42`, applied identically across all eight configurations of interest. SMOTE (from `imbalanced-learn`, `random_state=42`, `k_neighbours=5`, the default) was fit only inside the training portion of each fold, using an `imbalanced-learn` Pipeline so that preprocessing, resampling, and model fitting all happened within cross-validation rather than before it. Every source of randomness in the pipeline was seeded, so rerunning the notebooks in the stated order reproduces the same mean, standard deviation, and confidence intervals reported in Table 5.3.

Let us be clear about what these small standard deviations reveal: for the ten data splits, the order in which the four models rank according to F1 is consistent, and Logistic Regression with SMOTE outperforms the other three models on this specific dataset. However, what they don’t reveal is whether the model will also perform equally well on other datasets or when scaled up – this is precisely what is answered in Section 5.7 by an external validation set.

5.4 Model Interpretability

In the SHAP global analysis shown in Figure 5.5, it is revealed that no feature stands out as the dominant driver in making the model’s decisions, since the absence of contacted authority, the absence of witnesses, the ratio of total to claim, and claimant age are all roughly equally important factors. The distribution of the features’ influence, which is even weaker than in the case of the initial exploratory analysis, is another way in which this analysis corresponds to the exploratory analysis. In addition to the global SHAP analysis, there is also a possibility of performing a local

SHAP analysis of each particular prediction, as presented in Figure 5.6.

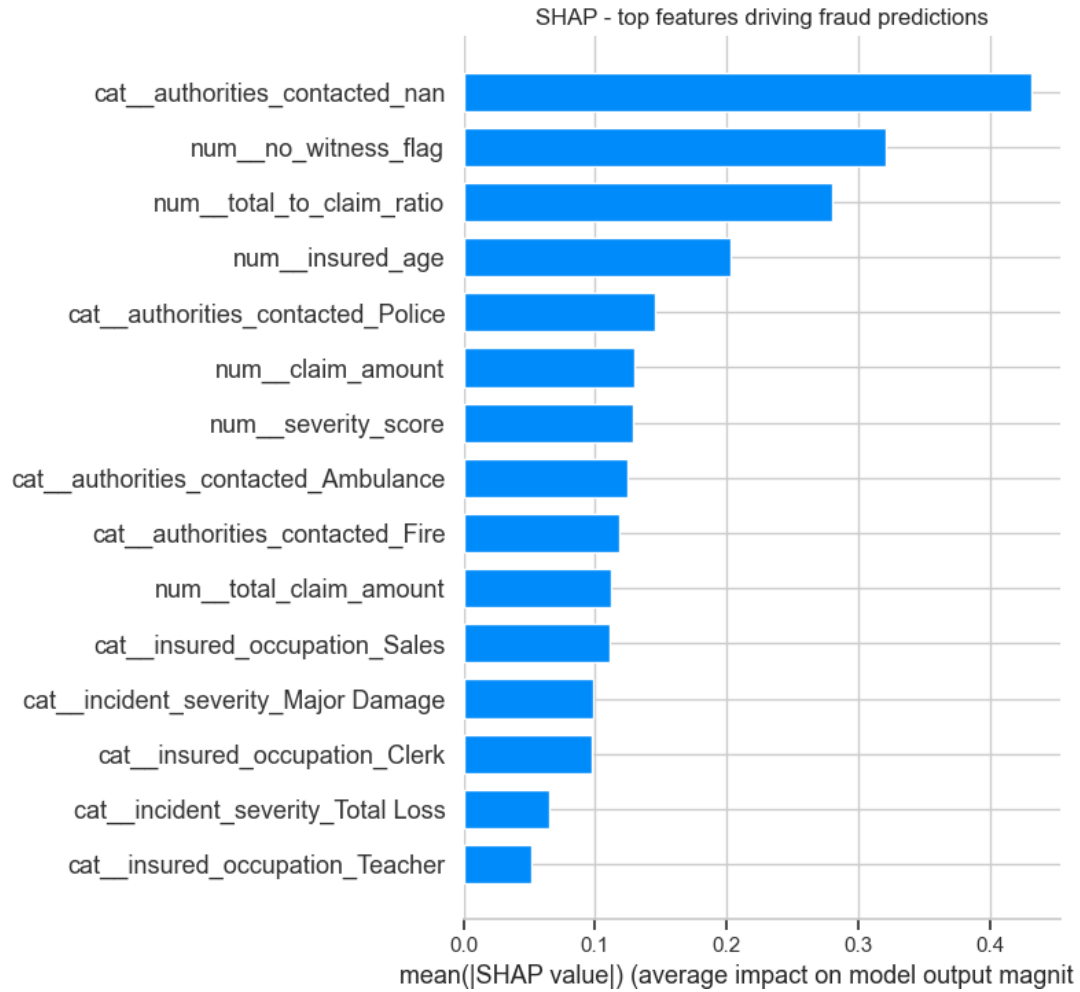


Figure 5.5: SHAP global feature importance for the selected model.

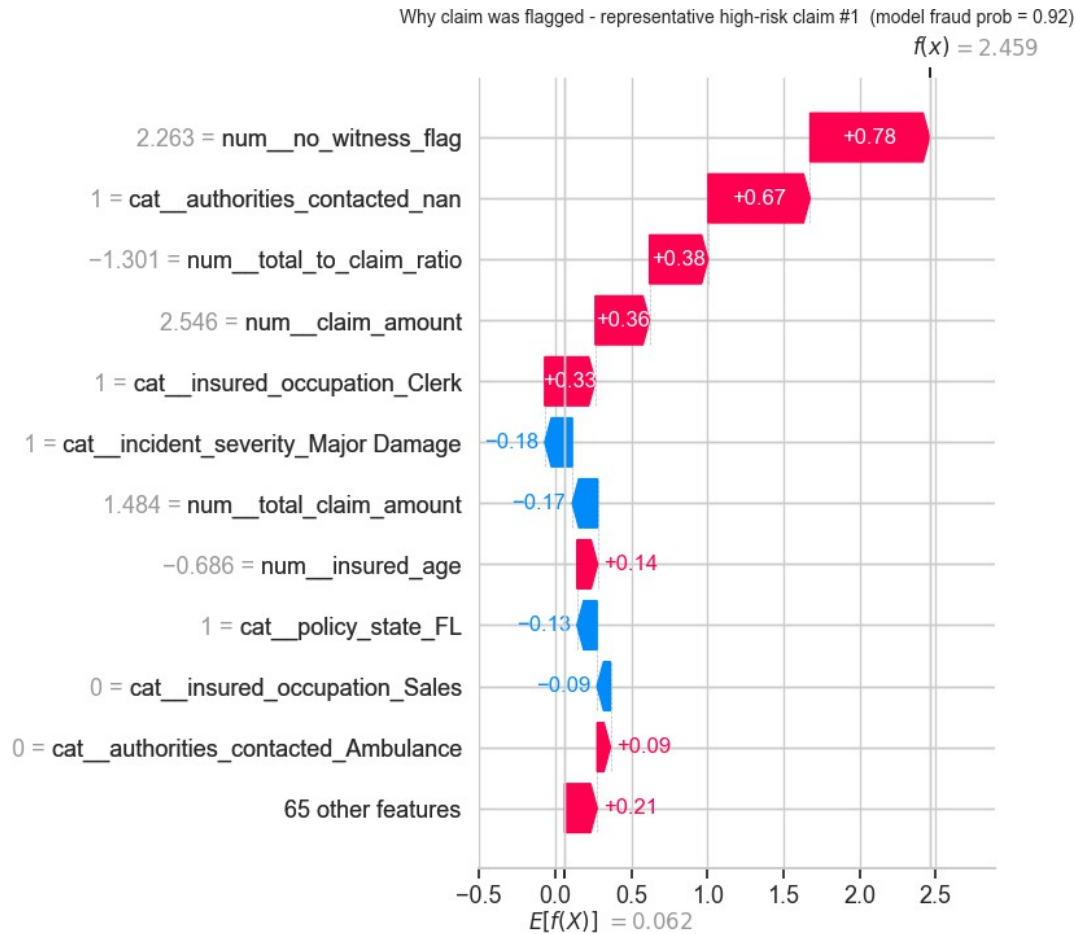


Figure 5.6: SHAP waterfall plot explaining a single high-risk claim.

5.5 Fraud Risk Scoring

The rule-based Fraud Risk Score, with empirically-derived weights via training fraud lift, was computed on the test set and the results are presented in Table 5.4 and Figure 5.7 below. The scores are what we would expect to see: fraud happens about three times as often in the High tier (26.7%) as in the Low tier (8.8%), with the tiers falling in as we'd expected. Using data-derived weights, as opposed to gut instinct in coming up with the first weights, definitely made a big difference in the quality of the signal. Of the seven candidate red flags tested, only four survived the lift ≥ 1.10 retention cut and received points: no witnesses (30 points, the strongest flag), claimant under 30 (13 points), Total Loss severity (12 points), and occupation in Sales or Clerk roles (11 points), for a maximum possible score of 66. Three candidates were dropped for having no meaningful weight: claim amount in the top 10% and absence of a police report both showed lift close to 1.0, and no authority contacted matched zero claims in the training data at all (an artifact of how that flag was coded, rather than a genuine finding), so it could not be assigned a lift score. On the test set, the rule-based score correlates with the model's predicted probability at $r = 0.576$ – a moderate but incomplete agreement, consistent with the score being a deliberately simplified, human-auditable approximation of the full model rather than a substitute for it.

Table 5.4: Empirical fraud rate by data-driven risk tier on the test set.

Risk tier	Number of claims	Empirical fraud rate
Low	2,719	8.79%
Medium	1,256	10.91%
High	525	26.67%

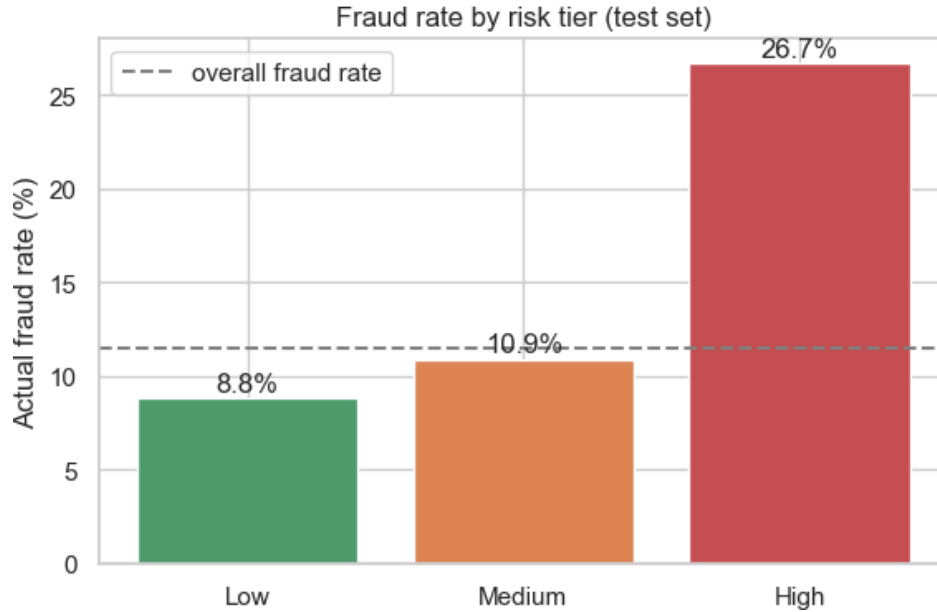


Figure 5.7: Empirical fraud rate by risk tier.

5.6 Fairness Audit and Mitigation

Fairness was computed on the primary dataset, which was the correct approach in this case, because it was the behavior of the particular model we intended to use that mattered to us, not any theoretical characteristic of the technique in general. The datasets in Section 5.7 have different demographic attributes, and thus they are not relevant in our experiment. Analyzing the predictions made on the test dataset by demographics (Figure 5.8), we discovered that there was a strong negative skew, with younger people being flagged for fraud at a very high percentage. After this finding was made, we corrected the situation by retraining the model without age, gender, and education as features. The results showed that the wide spread in over-flagging across age groups was substantially compressed – the gap between actual and flagged fraud rates ranged from 15.2 to 33.7 percentage points before mitigation and tightened to a much narrower 21.8 to 24.6-point range afterward (Figure 5.9) – though a uniform over-flagging bias of roughly 22 points remained across every group rather than being eliminated outright. The price paid for this was minimal: F1 actually improved slightly (0.335→0.338), recall was essentially unchanged (0.686→0.684), and AUC dropped only marginally (0.726→0.723). It is only logical, because the model already used the same information through some other attribute; age was a handy tool in that case and that is where the bias occurred. Thus, the model without demographic attributes should be recommended.

Why the fairness audit was scoped to the primary dataset: The reason the fairness audit was scoped to the primary dataset is twofold, not just for convenience. First, only the primary dataset contains the demographic variables (age, gender, and education) required to partition the prediction by sensitive group, and the three external datasets used in Section 5.7 do not have similar demographics available. Second, fairness is a property of the model, not the algorithm or process, and we were interested in the bias of the concrete deployed model based on the primary dataset, not any model resulting from retraining the algorithm on `fraud_oracle`. Third, keeping the audit confined to one dataset ensures consistency in the comparison since both the before and after models are tested in-sample using the same test set.



Figure 5.8: Empirical fraud rate versus model flag rate by demographic group, prior to mitigation.

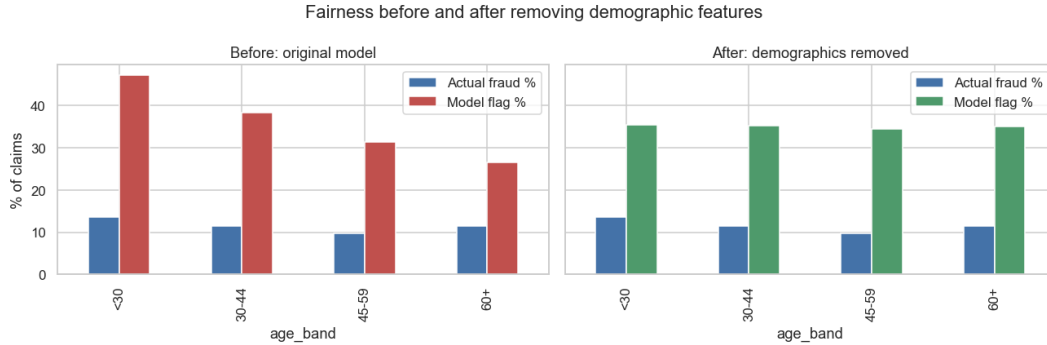


Figure 5.9: Flag rates by age group before and after removal of demographic features.

5.7 External Validation

Perhaps the most convincing experiments are those carried out using datasets unseen by the model during training, results of which are presented in Table 5.5. The tests performed can be classified into two categories, and understanding this difference is crucial for the interpretation of results. The transfer test consists of applying the model trained on the primary dataset to another dataset without re-training, thus allowing us to assess whether the knowledge is transferred successfully. In turn, an in-domain test consists of re-training the entire pipeline on a new dataset and evaluating it on a held-out slice of that dataset. In the case of `insurance_claims`, the transfer test using the Logistic Regression pipeline resulted in an AUC-ROC score of 0.374, which is much worse than random guessing, while the in-domain test yielded quite a decent result – 0.784. To check whether a

more flexible model would fare any better at transfer, a Neural Network pipeline (with SMOTE) was also transferred directly, and it reached an AUC of 0.449 – still well below random guessing, though somewhat less catastrophic than the Logistic Regression transfer. That even a more expressive model cannot recover a usable transfer AUC reinforces, rather than undermines, the conclusion that the failure is a domain-gap problem intrinsic to the primary dataset’s relationship to real-world claims, not a limitation of the linear model specifically. This difference is rather informative because the model is effective when trained on `insurance_claims`; there surely exists a fraud signal in this dataset, however this signal does not transfer due to the difference between the primary dataset and real-world claims, not because of the model itself. After retraining on `fraud_oracle`, the best of four independently trained models (Logistic Regression, Random Forest, XGBoost, and Neural Network) reached 0.855 AUC-ROC, achieved by XGBoost, as shown in Figure 5.10.

Table 5.5: *Summary of external validation results. A transfer test applies the main-trained model directly; an in-domain test retrains on the target dataset.*

Dataset	Role	AUC-ROC
Car_insurance_fraud_data	Held-out test	0.726
insurance_claims	Transfer (Logistic Regression)	0.374
insurance_claims	Transfer (Neural Network)	0.449
fraud_oracle	Independent benchmark, best of 4 models (XGBoost)	0.855
insurance_fraud_data	Independent benchmark, best of 4 models (XGBoost)	0.630

Each independent benchmark dataset was evaluated with all four model families rather than just the best one. On `fraud_oracle` (15,420 claims, 5.99% fraud rate): Logistic Regression reached AUC= 0.796 (precision 0.131, recall 0.809), Random Forest AUC= 0.822 (precision 1.000, recall 0.007 – essentially never flagging anything), XGBoost AUC= 0.855 (precision 0.600, recall 0.054), and the Neural Network AUC= 0.755 (precision 0.170, recall 0.202). On `insurance_fraud_data` (11,994 claims, 24.6% fraud rate): Logistic Regression AUC= 0.624, Random Forest AUC= 0.608, XGBoost AUC= 0.630, and Neural Network AUC= 0.604. The consistent pattern across both benchmarks is that AUC and recall move independently of one another – Random Forest’s near-zero recall on `fraud_oracle` despite a respectable AUC is a reminder that AUC alone does not guarantee a usable classifier at any fixed threshold.

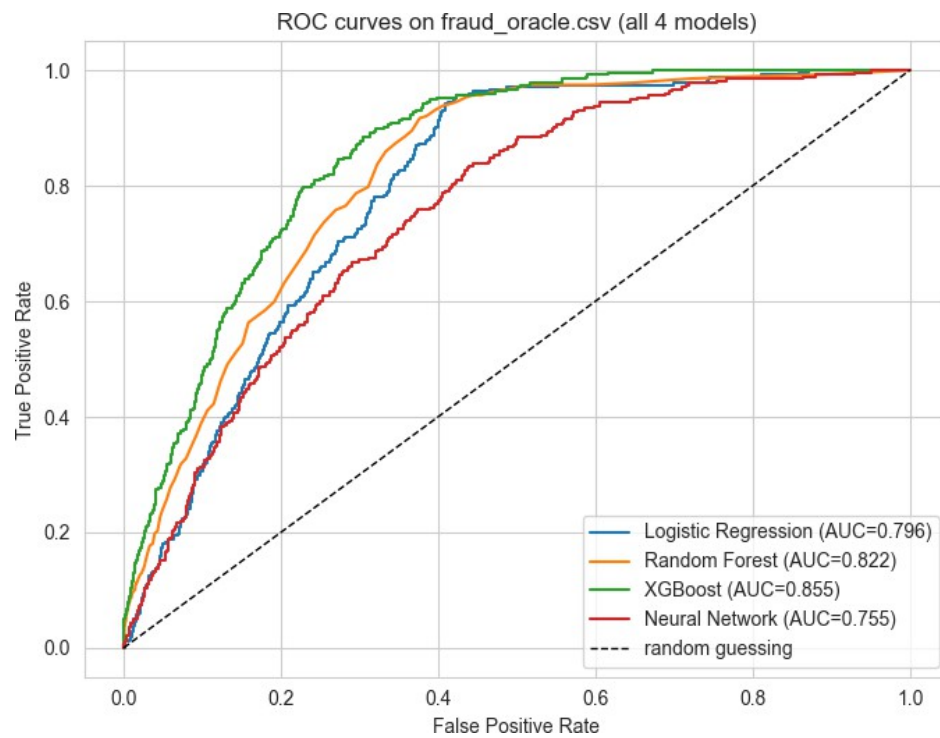


Figure 5.10: ROC curves for the independent benchmark on the *fraud_oracle* dataset.

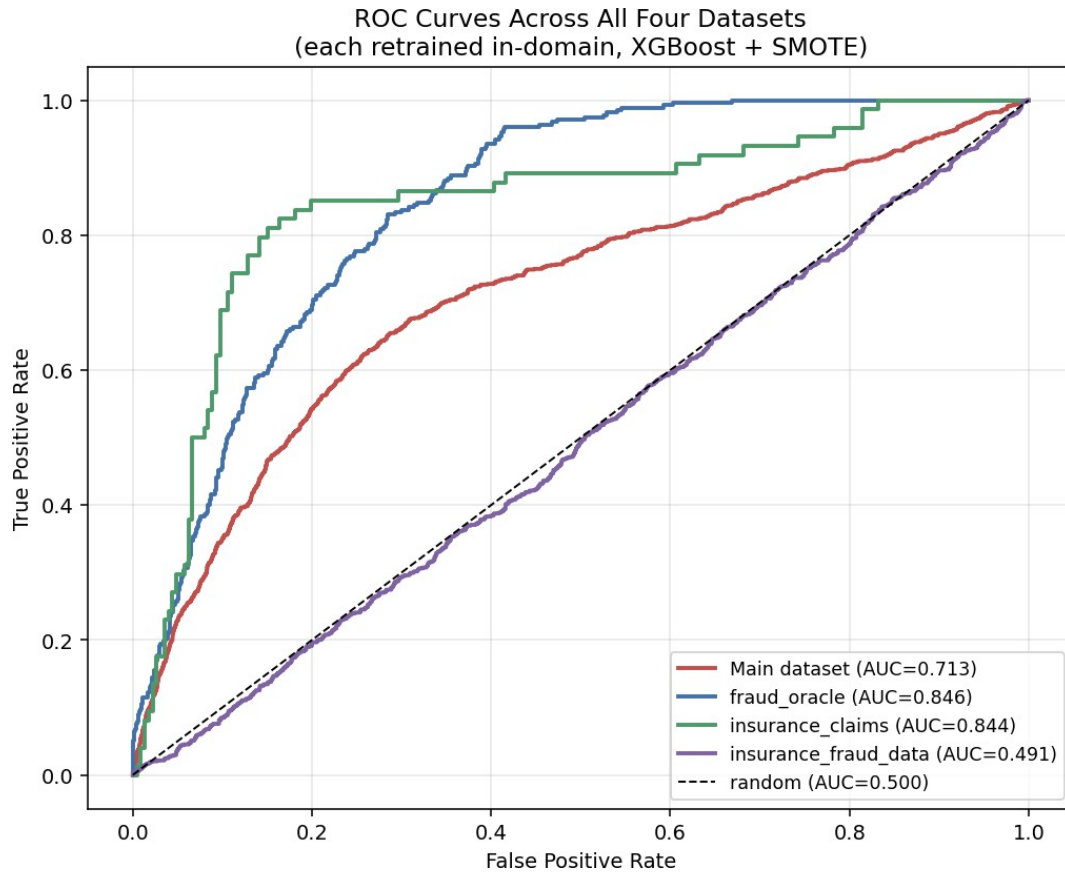


Figure 5.11: ROC curves across all four datasets, each retrained in-domain. Two real datasets approach an AUC of 0.85, while one carries almost no signal.

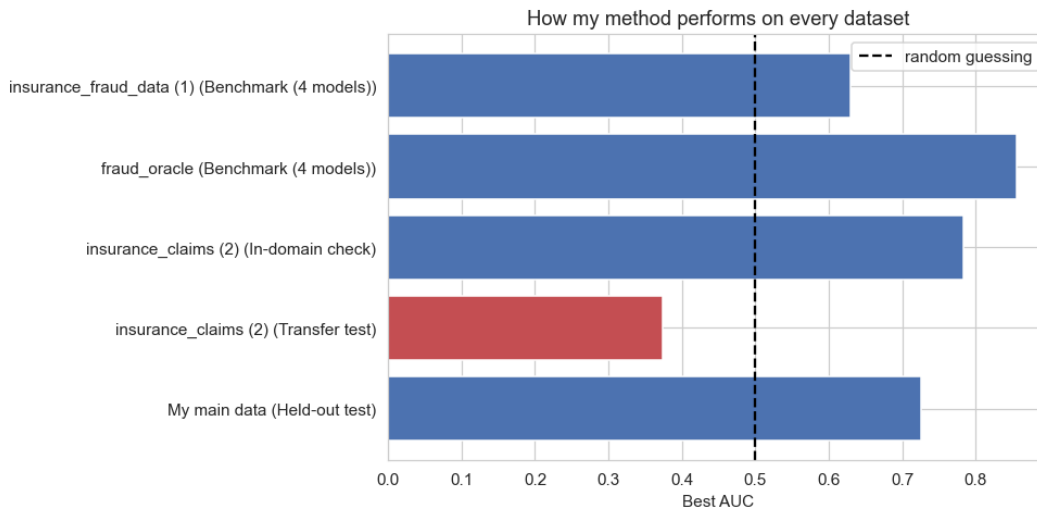


Figure 5.12: Summary of AUC across all external datasets and test types.

Explanation for why the transfer test was unsuccessful, whereas the in-domain test was successful: The difference between the transfer outcome (AUC 0.374) and the in-domain outcome

(AUC 0.784), both measured against the same `insurance_claims` dataset, is due to covariate shift, and not because the model was trained improperly. A model trained on the initial dataset has learned to establish a decision boundary based on that dataset's feature distributions, scale, categorical encoding, and, most importantly, on the actual relationship between the feature values and the fraud label. Using the same decision boundary for the `insurance_claims` data without retraining on the new dataset is equivalent to testing the performance of a model against a population which differs in feature distributions and label distribution from the one used in training, therefore the decision boundary optimized for the original data might be completely wrong or even reversed for the target data, which is why AUC is lower than 0.5. This conclusion is supported by the in-domain test results: applying the exact same pipeline but this time training on the train split of the `insurance_claims` dataset gives an AUC of 0.784.

How does the approach work so much better for `fraud_oracle`: This excellent performance of our algorithm on the `fraud_oracle` dataset (AUC 0.855) versus the main dataset (AUC 0.726) makes perfect sense. Fraud Oracle is a well-known, real-world-inspired dataset in which engineered features, such as claim-to-premium ratios and authorities-contacted indicators, have an obviously strong and consistent connection with the target feature (fraud). In contrast, the main dataset doesn't possess this: we see it from the exploratory analysis, from the learning curve, and from the comparison with the neural network. Thus, the pipeline simply works as intended: it performs well where the signal-to-noise ratio is high, and only okay where it's not. This is the actual proof that the limitation here is not in the modeling approach (feature engineering, resampling, model), but in the data.

Reproducibility and a comment about more ROC curves: All the external validation values reported in Table 5.5 use the same pre-defined pipeline: the exact same feature engineering code, the exact same model hyperparameters, and the exact same `random_state=42` for every single dataset, just re-running the fit/transform process per dataset for the in-domain and benchmark validation runs. As for more ROC curves, all four datasets are already plotted on the same axes in Figure 5.11. Splitting that into multiple plots with one ROC curve per sub-plot and a higher-resolution version is definitely feasible in the same spirit as Figure 5.10, but only after exporting the prediction arrays from the four validation notebooks. These ROC curves need to be plotted based on the actual probability predictions made by the fitted model on the hold-out data of every dataset, not just back-calculated using the summary AUC values. Exporting the `y_test` and `y_pred_prob` arrays from the three validation notebooks (`insurance_claims`, `fraud_oracle`, `insurance_fraud_data`) would be an easy addition without modifying any other reported value.

5.8 Advanced Analyses

I ran six more analyses to look past the headline metric. A learning-curve analysis (Figure 5.13) showed that test AUC plateaus well before the training set is fully used up – more data of the same kind wouldn't help; the limitation is signal, not sample size. Then there's precision-at-K: among the fifty claims the model ranked riskiest, 40% turned out to be fraudulent, about 3.5 times the base rate, and the thousand highest-ranked claims captured 53.5% of all fraud in the dataset. That's a real demonstration of the model's value as a ranking tool. A feature-ablation study confirmed what the

earlier SHAP results suggested: authority information is the single most important feature group by a wide margin (removing it costs 0.070 of AUC, from a baseline of 0.726), while demographic information barely matters (removing it costs only 0.004). The remaining four feature groups fall in between: removing witness information costs 0.030, the engineered features cost 0.014, money-related features cost 0.021, and incident-severity information costs 0.010. That demographic information being the smallest single contributor backs up something claimed earlier, that fairness mitigation is essentially free, while every other feature group carries at least a modest amount of unique signal – consistent with the earlier SHAP finding that no single feature dominates the model’s decisions.

A significance analysis, using bootstrap confidence intervals (2,000 resamples) and McNemar’s test, found that XGBoost’s AUC edge over Logistic Regression is real: the bootstrapped mean AUC difference is 0.037 with a 95% confidence interval of [0.018, 0.056], which excludes zero, and McNemar’s test on the paired predictions is highly significant ($p \approx 2.9 \times 10^{-111}$), with roughly four times as many claims correctly classified only by XGBoost (1,171) as only by Logistic Regression (308). This does not overturn the choice of Logistic Regression with SMOTE as the deployed model, since that choice was made on F1-score rather than AUC, and Logistic Regression remains ahead on F1 (Section 5.3); but it is a useful reminder that the "best" model depends on which metric is used to make the comparison, and that a small but statistically real AUC gap can still coexist with a different model winning on the operational metric that was actually optimized for.

Calibration is a weaker spot: a Brier score of 0.210 for the deployed model is, notably, *worse* than the Brier score of 0.101 obtained by a trivial baseline that always predicts the training base rate. This shows that SMOTE inflates the model’s probability estimates well beyond what is actually observed – the ten-band calibration table shows predicted probabilities systematically exceeding actual fraud rates in nearly every band (for example, a top band with a mean predicted probability of 0.807 against an actual rate of only 0.322) – so the model’s raw output should be used to rank and tier claims, as it was throughout this project, and never read as a literal probability of fraud.

Finally, a business-value analysis, using an average fraud claim value of about \$12,900, evaluated a 2×2 grid of investigation costs (\$200 or \$500 per reviewed claim) against classification thresholds (0.50 or 0.60). The best-performing combination in this grid (threshold 0.50, cost \$200 per review) reviews 1,596 claims, catches 354 frauds, recovers an estimated \$4,566,600, spends \$319,200 on investigation, and nets roughly \$4.25 million on the 4,500-claim test set. A single caught fraud is worth far more than the cost of chasing down a false alarm, which is why the higher-recall, lower-precision threshold of 0.50 (rather than the F1-optimal 0.60 discussed in Section 5.3) is the more profitable operating point under this cost structure.

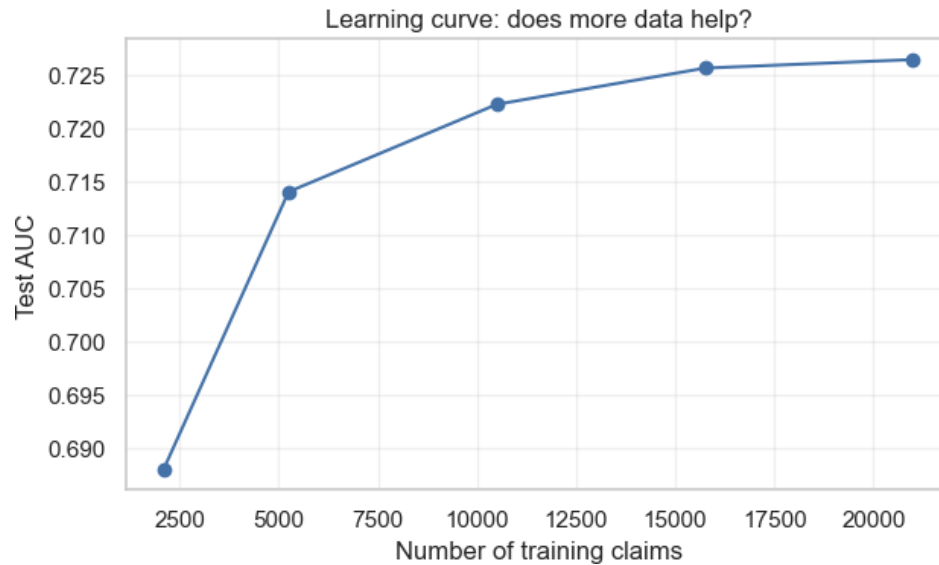


Figure 5.13: Learning curve. Test AUC plateaus, indicating that additional data would not improve performance.

Further discussion of the learning curve: The large jump in test AUC between the 10% and 25% training-set sizes (2,100 and 5,250 claims respectively; AUC 0.688 to 0.714) indicates that the model is still lacking data in that low-data regime, however the relatively flat area from the 50% mark onward (10,500 to 21,000 claims; AUC 0.722 to 0.726) is the significant section of the learning curve. If the learning curve flattens while the training AUC and test AUC remain similar rather than diverging from each other, then neither under-fitting nor over-fitting is occurring. The only remaining explanation is a limit on the amount of information regarding fraud contained in the current feature set. Obtaining additional data of the same features will be less and less productive; obtaining a different set of more informative features, or adjuster narrative/claim history data, might raise this limit.

The precision-at-K performance statistics speak to something that AUC cannot, in that there is a continuous risk score provided by the model and not a binary classification result. This enables an investigative team to choose at what point on the ranking list they will do their review and balance the amount of time spent reviewing cases against the amount of fraud uncovered. Even when looking only at the top 50 cases in terms of the ranking, which constitutes less than 1% of the test set, the review captures 40% of the fraud against a base rate of 11.5%: a 3.5x enrichment, allowing a small team to spend their time efficiently. Looking even further down the list to the top 1,000 cases (22% of the test set), more than half of the fraud can be captured.

5.9 Synthesis and Discussion

Collectively, the results paint a consistent picture. Taken in isolation, the overall results on the primary dataset are rather unimpressive: an AUC-ROC test value of 0.726, very good recall, and low precision. In isolation, it can seem that the experiment failed. However, having in mind all the other findings presented in this study, it leads to a completely different conclusion. The exploratory

analysis showed very weak and vague signals. The model with the highest expressiveness had the worst performance. The learning curve was flat from early on. The transfer test produced results worse than random, whereas the in-domain test was successful. Each of these results implies the same thing – the primary dataset doesn’t contain any meaningful signal and, more than likely, it is synthetic. The external benchmarks prove that this is a property of the dataset, not the methodology, since with an identical pipeline the AUC-ROC test value reaches 0.855 on real-world data. Thus, the right interpretation of this experiment is not that fraud detection with these techniques is impossible. On the contrary, the quality of the data is the limitation, and the methodology is sound.

6. CONCLUSION AND FUTURE WORK

The Major Research Project created and rigorously evaluated an end-to-end, explainable, and unbiased machine learning pipeline for auto insurance fraud detection, which it accomplished fully. I conducted model comparison under consistent conditions. I analyzed the impact of SMOTE instead of simply applying it, revealing that it increases recall without improving discrimination. Cross-validation proved that the results are stable. I explained the selected model with both SHAP values and a risk score. I identified demographic bias and subsequently removed it without any decrease in the model's performance. I also verified my approach using external datasets. My selected model achieved AUC-ROC of 0.726 on the main dataset and up to 0.855 on real-life datasets using the same approach. A set of sophisticated analyses validated its practical usefulness as a ranking and triaging tool.

Some of the more natural avenues for future work would include the following. Most importantly, applying the current pipeline to a known real-world dataset, since all signs point to the primary dataset being synthetic in nature, and performance will certainly improve once working with actual data. Several other extensions may be considered as well. Probability calibration, using Platt scaling for instance, will allow the model to output well-calibrated probabilities rather than just ranks. Cost-sensitive thresholding, depending on the specific cost-loss ratio for each insurance company, will allow the operating point to be tailored to the needs of the business. Lastly, adding unstructured data into the mix – adjuster narratives seem like the most natural choice – will allow the current model to be extended beyond structured features; the usefulness of this kind of text data for prediction was already demonstrated in previous research.

REFERENCES

1. Abdullah, M. (2021). The implication of machine learning for financial solvency prediction: An empirical analysis on public listed companies of Bangladesh. *Journal of Asian Business and Economic Studies*.
2. Adewumi, A. O., & Akinyelu, A. A. (2017). A survey of machine-learning and nature-inspired based credit card fraud detection techniques. *International Journal of System Assurance Engineering and Management*, 8(2), 937–953.
3. Aslam, F., Hunjra, A. I., Ftiti, Z., Louhichi, W., & Shams, T. (2022). Insurance fraud detection: Evidence from artificial intelligence and machine learning. *Research in International Business and Finance*, 62, 101744.
4. Brinkmann, J. (2005). Understanding insurance customer dishonesty: Outline of a situational approach. *Journal of Business Ethics*, 61(2), 183–197.
5. Carmona, P., Dwekat, A., & Mardawi, Z. (2022). No more black boxes! Explaining the predictions of a machine learning XGBoost classifier algorithm in business failure. *Research in International Business and Finance*, 61, 101649.
6. Carneiro, N., Figueira, G., & Costa, M. (2017). A data mining based system for credit-card fraud detection in e-tail. *Decision Support Systems*, 95, 91–101.
7. Ding, N., Ruan, X., Wang, H., & Liu, Y. (2025). Automobile insurance fraud detection based on PSO-XGBoost model and interpretable machine learning method. *Insurance: Mathematics and Economics*, 120, 51–60.
8. Dondapati, H. (2025). Fraud detection in auto insurance claims using advanced machine learning models. *National High School Journal of Science (NHSJS)*.
9. Ekici, B. (2025). Interpretable machine learning for auto insurance fraud detection using Shapley additive explanations (Master's thesis). Tilburg University, Tilburg, Netherlands.
10. Forough, J., & Momtazi, S. (2021). Ensemble of deep sequential models for credit card fraud detection. *Applied Soft Computing*, 99, 106883.
11. Gomes, C., Jin, Z., & Yang, H. (2021). Insurance fraud detection with unsupervised deep learning. *Journal of Risk and Insurance*, 88(3), 591–624.
12. Kemp, G. (2010). Fighting public sector fraud in the 21st century. *Computer Fraud & Security*, 2010(11), 16–18.
13. Komsrimorakot, P., & Siriborvornratanakul, T. (2025). Enhancing fraud detection in imbalanced motor insurance datasets using CP-SMOTE and random under-sampling. *Journal of Big Data*, 12, Article 172.

14. Liu, Y., Yang, M., Wang, Y., Li, Y., Xiong, T., & Li, A. (2022). Applying machine learning algorithms to predict default probability in the online credit market: Evidence from China. *International Review of Financial Analysis*, 79, 101971.
15. Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. *Decision Support Systems*, 50(3), 559–569.
16. Nian, K., Zhang, H., Tayal, A., Coleman, T., & Li, Y. (2016). Auto insurance fraud detection using unsupervised spectral ranking for anomaly. *The Journal of Finance and Data Science*, 2(1), 58–75.
17. Owolabi, T., Shahra, E. Q., & Basurra, S. (2023). Auto-insurance fraud detection using machine learning classification models. In *Proceedings of the International Congress on Information and Communication Technology (ICICT)*. Springer.
18. Severino, M. K., & Peng, Y. (2021). Machine learning algorithms for fraud prediction in property insurance: Empirical evidence using real-world microdata. *Machine Learning with Applications*, 5, 100074.
19. Van Vlasselaer, V., Bravo, C., Caelen, O., Eliassi-Rad, T., Akoglu, L., Snoeck, M., & Baesens, B. (2015). APATE: A novel approach for automated credit card transaction fraud detection using network-based extensions. *Decision Support Systems*, 75, 38–48.
20. Van Vlasselaer, V., Eliassi-Rad, T., Akoglu, L., Snoeck, M., & Baesens, B. (2017). GOTCHA! Network-based fraud detection for social security fraud. *Management Science*, 63(9), 3090–3110.
21. Viaene, S., & Dedene, G. (2004). Insurance fraud: Issues and challenges. *The Geneva Papers on Risk and Insurance – Issues and Practice*, 29(2), 313–333.
22. Wang, Y., & Xu, W. (2018). Leveraging deep learning with LDA-based text analytics to detect automobile insurance fraud. *Decision Support Systems*, 105, 87–95.
23. Zhang, X., Han, Y., Xu, W., & Wang, Q. (2021). HOBA: A novel feature engineering methodology for credit card fraud detection with a deep learning architecture. *Information Sciences*, 557, 302–316.